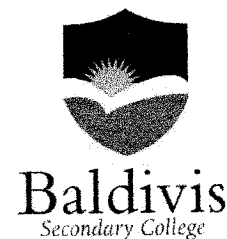


Year 10

Investigation 1



Surface Area and Volume

SOLUTIONS

Total Mark: _____

Name: _____ Class: _____ Due Date : _____

This investigation has two parts- TAKE HOME component and IN-CLASS VALIDATION.

Allocated marks

TAKE HOME: _____/20 Marks

IN-CLASS VALIDATION: _____/37 Marks

TOTAL: _____/

Total: 57 Marks

***** START OF AT HOME SECTION*****

Introduction

In this investigation you are going to work through several calculations and designs to prove or disprove the following statement:

STATEMENT #1:

"The total surface area of a cube is always greater than its volume".

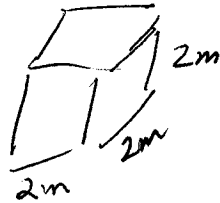
The following calculations will help you to collect evidence to make an informed decision on whether the statement above is correct or not.

Make sure you spend time working through these calculations as you will be able to use them to help you set out and answer the in-class validation section

You will do the Shape 1 calculations and evidence in class, with guidance provided by your teacher. It is important that you ask questions and follow the working out discussed so you are able to attain as many of the marks as possible for each question.

Shape 1 calculations and evidence

Sketch and label a cube with the dimensions of 2m wide, 2m long and 2m high



Calculate the surface area for this cube?

$$\begin{aligned} S.A &= (L \times l) \times 6 \\ &= 2 \times 2 \times 6 \\ &= 24 \text{ m}^2 \end{aligned}$$

Calculate the volume for this cube?

$$\begin{aligned} V &= L \times l \times h \\ &= 2 \times 2 \times 2 \\ &= 8 \text{ m}^3 \end{aligned}$$

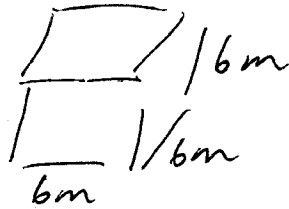
Looking back at your surface area and volume calculations for this 2m x 2m x 2m cube, write 1-2 sentences proving or disproving the statement.

In a 2x2x2 cube the surface area is greater than the volume.

Shape 2 calculations and evidence

Sketch and label a cube with the dimensions of a 6m x 6m x 6m

[1 mark]



$\frac{1}{2}$ mark for sketch

$\frac{1}{2}$ mark for dimensions

Calculate the surface area for this cube?

[3 marks]

$$\begin{aligned} S.A &= (L \times L) \times 6 \\ &= 6 \times 6 \times 6 \\ &= 216 \text{ m}^2 \end{aligned}$$

1 mark for formula
1 mark for substitution
1 mark for answer with correct dimensions.

Calculate the volume for this cube?

[3 marks]

$$\begin{aligned} V &= L \times L \times h \\ &= 6 \times 6 \times 6 \\ &= 216 \text{ m}^3 \end{aligned}$$

1 mark formula
1 mark substitution
1 mark answer with correct units.

Looking back at your surface area and volume calculations for this 6m x 6m x 6m cube, write 1-2 sentences proving or disproving the statement.

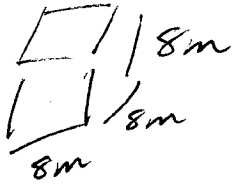
[1 mark]

The surface area + volume for a 6x6x6 cube are equal. Surface area is 216 m^2 + its volume is 216 m^3 .

Shape 3 calculations and evidence

Sketch and label a cube with the dimensions of a 8m x 8m x 8m

[1 mark]



See mark breakdown
from Shape 2.

Calculate the surface area for this cube?

[3 marks]

$$\begin{aligned} S.A &= (L \times L) \times 6 \\ &= 8 \times 8 \times 6 \\ &= 384m^2 \end{aligned}$$

See mark breakdown
from Shape 2.

Calculate the volume for this cube?

[3 marks]

$$\begin{aligned} V &= L \times L \times h \\ &= 8 \times 8 \times 8 \\ &= 512m^3 \end{aligned}$$

See mark breakdown
from Shape 2.

Looking back at your surface area and volume calculations for this 8m x 8m x 8m cube, write 1-2 sentences proving or disproving the statement.

[1 mark]

For an 8x8x8 cube the volume is
larger ^① than the surface area

Example :

Your summary should include:

- What you are answering
- Re-stating your calculations
- What did you find out?
- What is your opinion of the statement

Surface area + volume calculations were found for 3 different sized cubes. (1)

For the $2 \times 2 \times 2$ cube the S.A was 24m^2 + its volume was 8m^3 . The $6 \times 6 \times 6$ cube had a S.A of 216m^2 and its volume was 216m^3 . Finally, the largest cube of $8 \times 8 \times 8$ had a S.A of 384m^2 + a volume of 512m^3 . (1)

The smallest cube demonstrated that the S.A was larger than the volume. For the $6 \times 6 \times 6$ cube the S.A + volume were exactly the same. Once you have a cube larger than a $6 \times 6 \times 6$ cube, the volume becomes larger than its surface area. The $8 \times 8 \times 8$ demonstrated this. (1)

My calculations disproved statement #1, as cubes that have dimensions greater than $6 \times 6 \times 6$ have volumes larger than their surface areas (1)

*** END OF AT HOME SECTION***